CHAPTER 26

The Urinary System: Functional Anatomy and Urine Formation By the Kidneys

MULTIPLE FUNCTIONS OF THE KIDNEYS

Most people are familiar with one important function of the kidneys—to rid the body of waste materials that are ingested or produced by metabolism. A second function that is especially critical is to control the volume and electrolyte composition of the body fluids. For water and virtually all electrolytes in the body, the balance between intake (due to ingestion or metabolic production) and output (due to excretion or metabolic consumption) is maintained largely by the kidneys. This regulatory function of the kidneys is essential for maintaining the stable internal environment necessary for the cells to perform their various activities.

The kidneys perform many of their most critical functions by filtering the plasma and removing substances from the filtrate at variable rates, depending on the needs of the body. Ultimately, the kidneys clear unwanted substances from the filtrate (and therefore from the blood plasma) by excreting them in the urine while returning substances that are needed back to the blood.

Although this chapter and the next few chapters focus mainly on the control of renal excretion of water, electrolytes, and metabolic waste products, the kidneys serve many important homeostatic functions, including the following:

- Excretion of metabolic waste products and foreign chemicals
- Regulation of water and electrolyte balances
- Regulation of body fluid osmolality and electrolyte concentrations
- Regulation of arterial pressure
- Regulation of acid–base balance
- Regulation of erythrocyte production
- Secretion, metabolism, and excretion of hormones
- Gluconeogenesis

Excretion of Metabolic Waste Products, Foreign Chemicals, Drugs, and Hormone Metabolites. The kidneys are the primary means for eliminating most of the waste products of metabolism that are no longer needed by the body. These products include *urea* (from the metabolism of amino acids), *creatinine* (from muscle creatine), *uric acid* (from nucleic acids), *end products of hemoglobin breakdown* (e.g., bilirubin), and *metabolites of various hormones*. These waste products are usually eliminated from the body as rapidly as they are produced. The kidneys also eliminate most toxins and other foreign substances that are produced by the body or ingested, such as pesticides, drugs, and food additives.

Regulation of Water and Electrolyte Balances. For maintenance of homeostasis, excretion of water and electrolytes must, in the long term, match intake precisely. If intake exceeds excretion, the amount of that substance in the body will increase. If intake is less than excretion, the amount of that substance in the body will decrease. Although temporary (or cyclic) imbalances of water and electrolytes may occur in various physiological and pathophysiological conditions associated with altered intake or renal excretion, the maintenance of life depends on restoration of water and electrolyte balance.

Intake of water and many electrolytes is usually governed by a person's eating and drinking habits, requiring the kidneys to adjust their excretion rates to match the intakes of various substances. **Fig. 26.1** shows the response of the kidneys to a sudden 10-fold increase in sodium intake from a low level of 30 mEq/day to a high level of 300 mEq/day. Within 2 to 3 days after raising the sodium intake, renal excretion also increases to about 300 mEq/day so that the balance between intake and output is rapidly reestablished. However, during the 2 to 3 days of renal adaptation to the high sodium intake, there is usually a modest accumulation of sodium that raises extracellular fluid volume slightly and triggers hormonal changes and other compensatory responses that signal the kidneys to increase their sodium excretion.

The capability of the kidneys to alter sodium excretion in response to changes in sodium intake is tremendous. Experimental studies have shown that in many people, sodium intake can be increased to 1500 mEq/day (more than 10 times normal) or decreased to 10 mEq/day (<0.1 of normal), with relatively small changes in extracellular fluid volume or plasma sodium concentration. This phenomenon is also true for water and for most other electrolytes, such as chloride, potassium, calcium, hydrogen, magnesium, and phosphate ions. In the next few chapters, we discuss the specific mechanisms that permit the kidneys to perform these amazing feats of homeostasis.

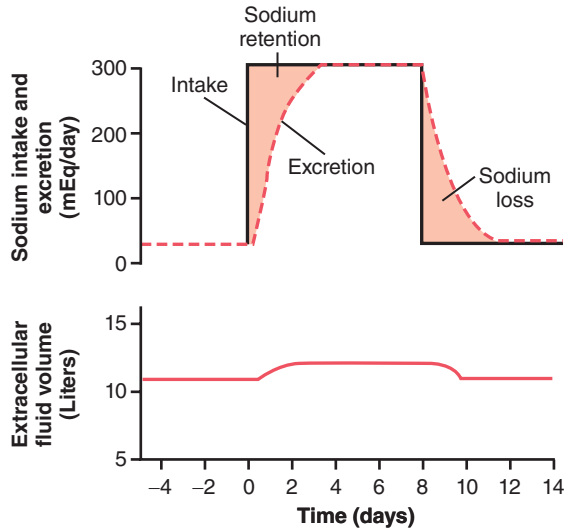


Figure 26.1 Effect of increasing sodium intake 10-fold (from 30 to 300 mEq/day) on urinary sodium excretion and extracellular fluid volume. The *shaded areas* represent the net sodium retention or net sodium loss, determined by the difference between sodium intake and sodium excretion.

Regulation of Arterial Pressure. As discussed in Chapters 19 and 30, the kidneys play a dominant role in long-term regulation of arterial pressure by excreting variable amounts of sodium and water. The kidneys also contribute to short-term arterial pressure regulation by secreting hormones and vasoactive factors or substances (e.g., *renin*) that lead to the formation of vasoactive products (e.g., angiotensin II).

Regulation of Acid–Base Balance. The kidneys contribute to acid–base regulation, along with the lungs and body fluid buffers, by excreting acids and by regulating the body fluid buffer stores, as discussed in Chapter 31. The kidneys are the only means of eliminating certain types of acids from the body, such as sulfuric acid and phosphoric acid, which are generated by the metabolism of proteins.

Regulation of Erythrocyte Production. The kidneys secrete *erythropoietin*, which stimulates production of red blood cells by *hematopoietic stem cells* in the bone marrow, as discussed in Chapter 33. One important stimulus for erythropoietin secretion by the kidneys is *hypoxia*. The kidneys normally account for almost all the erythropoietin secreted into the circulation. In people with severe kidney disease or who have had their kidneys removed and have been placed on hemodialysis, severe anemia develops as a result of decreased erythropoietin production.

Regulation of 1,25-Dihydroxyvitamin D_3 Production. The kidneys produce 1,25-dihydroxyvitamin D_3 (*calcitriol*), the active form of vitamin D, by hydroxylating this vitamin at the "number 1" position. Calcitriol is essential for normal calcium deposition in bone and calcium reabsorption by the gastrointestinal tract. As discussed in Chapter 80, calcitriol plays an important role in calcium and phosphate regulation.

Glucose Synthesis. The kidneys synthesize glucose from amino acids and other precursors during prolonged fasting, a process referred to as *gluconeogenesis*. The kidneys' capacity to add glucose to the blood during prolonged periods of fasting rivals that of the liver.

With chronic kidney disease or acute failure of the kidneys, these homeostatic functions are disrupted, and severe abnormalities of body fluid volumes and composition rapidly occur. With complete renal failure, enough potassium, acids, fluid, and other substances accumulate in the body to cause death within a few days unless clinical interventions such as hemodialysis are initiated to restore, at least partially, the body fluid and electrolyte balances.

PHYSIOLOGICAL ANATOMY OF THE KIDNEYS

GENERAL ORGANIZATION OF THE KIDNEYS AND URINARY TRACT

The two kidneys lie on the posterior wall of the abdomen, outside the peritoneal cavity (**Fig. 26.2**). Each kidney of the adult human weighs approximately 150 grams and is about the size of a clenched fist. Although kidney weight is generally greater in men than in women, this difference is partly related to greater body size in men. The medial side of each kidney contains an indented region called the *hilum*, through which pass the renal artery and vein, lymphatics, nerve supply, and ureter. The ureter carries the final urine from the kidney to the bladder, where it is stored until the bladder is emptied. The kidney is surrounded by a tough fibrous *capsule* that protects its delicate inner structures.

If the kidney is bisected from top to bottom, the two major regions that can be visualized are the outer *cortex* and the inner *medulla* regions. The medulla is divided into 8 to 10 cone-shaped masses of tissue called *renal pyramids*. The base of each pyramid originates at the border between the cortex and medulla and terminates in the *papilla*, which projects into the space of the *renal pelvis*, a funnel-shaped continuation of the upper end of the ureter. The outer border of the pelvis is divided into open-ended pouches called *major calyces* that extend downward and divide into *minor calyces*, which collect urine from the tubules of each papilla. The walls of the calyces, pelvis, and ureter contain contractile elements, including smooth muscle, that propel the urine toward the *bladder*, where urine is stored until it is emptied by *micturition*, discussed later in this chapter.

RENAL BLOOD SUPPLY

Blood flow to the two kidneys is normally about 22% of the cardiac output, or 1100 mL/min in the average adult, although considerable variation occurs, depending on various factors such as body size, age, and sex of the individual. The renal artery enters the kidney through

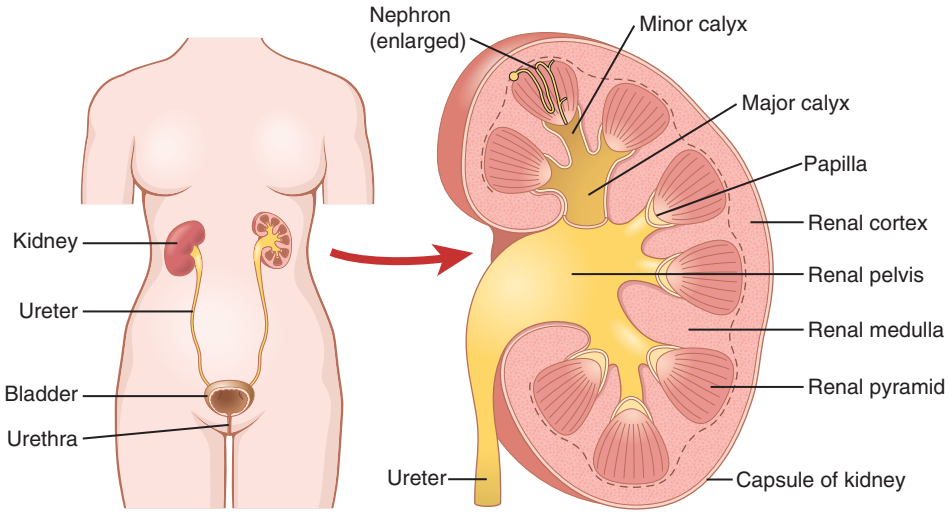


Figure 26.2 General organization of the kidneys and the urinary system.

the hilum and then branches progressively to form the *interlobar arteries, arcuate arteries, interlobular arteries* (also called *radial arteries*), and *afferent arterioles*, which lead to the *glomerular capillaries*, where large amounts of fluid and solutes (except the plasma proteins) are filtered to begin urine formation (**Fig. 26.3**). The distal ends of the capillaries of each glomerulus coalesce to form the *efferent arteriole*, which leads to a second capillary network, the *peritubular capillaries*, that surrounds the renal tubules.

The renal circulation is unique in having two capillary beds, the glomerular and peritubular capillaries, which are arranged in series and are separated by the efferent arterioles. These arterioles help regulate the hydrostatic pressure in both sets of capillaries. High hydrostatic pressure in the glomerular capillaries (~60 mm Hg) causes rapid fluid filtration, whereas a much lower hydrostatic pressure in the peritubular capillaries (~13 mm Hg) permits rapid fluid reabsorption. By adjusting the resistance of the afferent and efferent arterioles, the kidneys can regulate the hydrostatic pressure in the glomerular and the peritubular capillaries, thereby changing the rate of glomerular filtration, tubular reabsorption, or both in response to the body's homeostatic demands.

The peritubular capillaries empty into the vessels of the venous system, which run parallel to the arteriolar vessels. The blood vessels of the venous system progressively form the *interlobular vein, arcuate vein, interlobar vein*, and *renal vein*, which leaves the kidney beside the renal artery and ureter.

THE NEPHRON IS THE FUNCTIONAL UNIT OF THE KIDNEY

Each human kidney contains about 800,000 to 1,000,000 *nephrons*, each of which is capable of forming urine. The kidney cannot regenerate new nephrons. Therefore, with renal injury, disease, or normal aging, the number of nephrons gradually decreases. After age 40 years, the number of functioning nephrons usually decreases about 10% every 10 years; thus, at age 80 years, many people have 40% fewer functioning nephrons than they did at age 40 years. This loss is not life-threatening because adaptive changes in the remaining nephrons allow them to excrete the proper amounts of water, electrolytes, and waste products, as discussed in Chapter 32. However, common disorders such as obesity, hypertension, and diabetes mellitus may cause more rapid loss of nephrons with aging.

Each nephron contains (1) a tuft of glomerular capillaries called the *glomerulus*, through which large amounts of fluid are filtered from the blood, and (2) a long *tubule* in which the filtered fluid is converted into urine on its way to the pelvis of the kidney (see **Fig. 26.3**).

The glomerulus contains a network of branching and anastomosing glomerular capillaries that, compared with other capillaries, have high hydrostatic pressure (~60 mm Hg). The glomerular capillaries are covered by epithelial cells, and the total glomerulus is encased in *Bowman's capsule*.

Fluid filtered from the glomerular capillaries flows into Bowman's capsule and then into the *proximal tubule*, which lies in the cortex of the kidney (**Fig. 26.4**). From the proximal tubule, fluid flows into the *loop of Henle*, which dips into the renal medulla. Each loop consists of a *descending* and an *ascending limb*. The walls of the descending limb and lower end of the ascending limb are very thin and therefore are called the *thin segment of the loop of Henle*. After the ascending limb of the loop returns partway back to the cortex, its wall becomes much thicker; this segment is referred to as the *thick segment of the ascending limb*.

At the end of the thick ascending limb is a short segment that has in its wall a plaque of specialized epithelial cells, known as the *macula densa*. As discussed in Chapter 27, the macula densa plays an important role in controlling nephron function. Beyond the macula densa, fluid enters the *distal tubule*, which, like the proximal tubule, lies in the renal cortex. The distal tubule is followed by

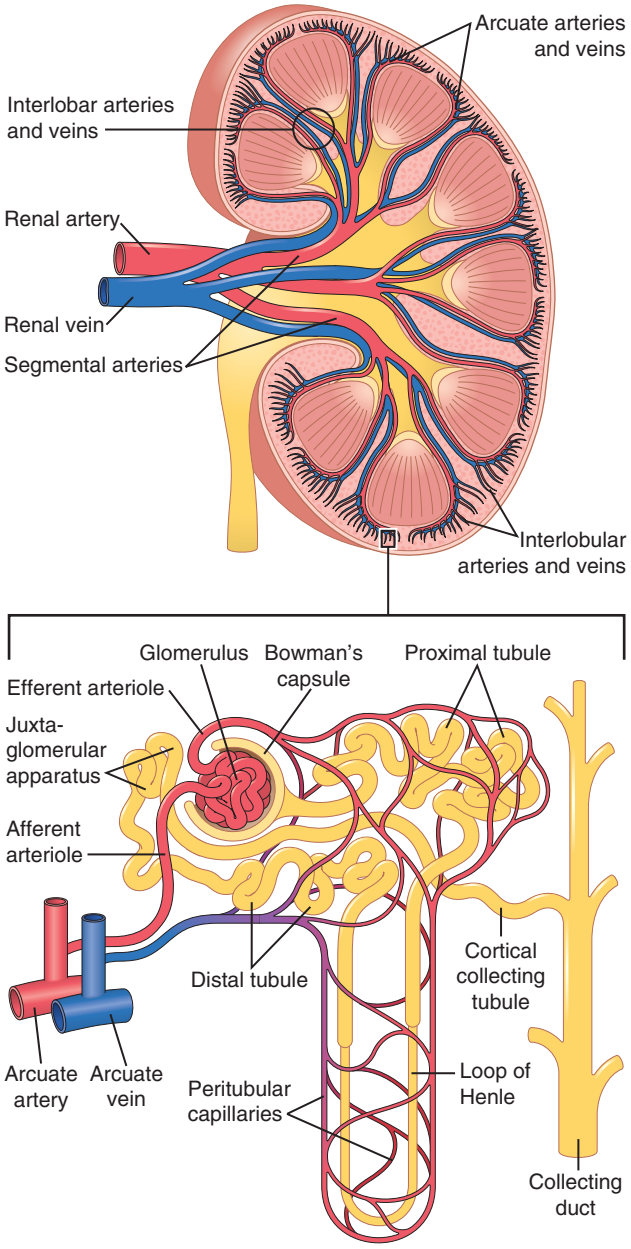


Figure 26.3 Section of the human kidney showing the major vessels that supply the blood flow to the kidney and a schematic of the microcirculation of each nephron.

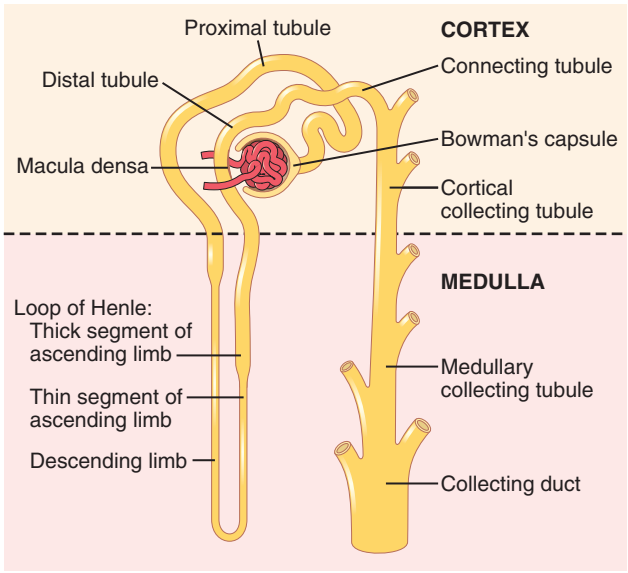


Figure 26.4 Basic tubular segments of the nephron. The relative lengths of the different tubular segments are not drawn to scale.

the *connecting tubule* and *cortical collecting tubule*, which lead to the *cortical collecting duct*. The initial parts of 8 to 10 cortical collecting ducts join to form a single, larger collecting duct that runs downward into the medulla and becomes the *medullary collecting duct*. The collecting ducts merge to form progressively larger ducts that eventually empty into the renal pelvis through the tips of the *renal papillae*. In each kidney, there are about 250 of these very large collecting ducts, each of which collects urine from about 4000 nephrons.

Regional Differences in Nephron Structure: Cortical and Juxtamedullary Nephrons. Although each nephron has all the components described earlier, there are some differences, depending on how deep the nephron lies within the kidney mass. The nephrons that have glomeruli located in the outer cortex are called *cortical nephrons*; they have short loops of Henle that penetrate only a short distance into the medulla (**Fig. 26.5**).

About 20% to 30% of the nephrons have glomeruli that lie deep in the renal cortex near the medulla and are called *juxtamedullary nephrons*. These nephrons have long loops of Henle that dip deeply into the medulla, in some cases all the way to the tips of the renal papillae.

The vascular structures supplying the juxtamedullary nephrons also differ from those supplying the cortical nephrons. For the cortical nephrons, the entire tubular system is surrounded by an extensive network of peritubular capillaries. For the juxtamedullary nephrons, long efferent arterioles extend from the glomeruli down into the outer medulla and then divide into specialized peritubular capillaries called *vasa recta*, which extend downward into the medulla, lying side by side with the loops of Henle. Like the loops of Henle, the vasa recta return toward the cortex and empty into the cortical veins. This specialized network of capillaries in the medulla plays an essential role in the formation of a concentrated urine, discussed in Chapter 29.

URINE FORMATION RESULTS FROM GLOMERULAR FILTRATION, TUBULAR REABSORPTION, AND TUBULAR SECRETION

The rates at which different substances are excreted in the urine represent the sum of three renal processes, shown in **Fig. 26.6**: (1) glomerular filtration of substances in the blood; (2) reabsorption of substances from the renal

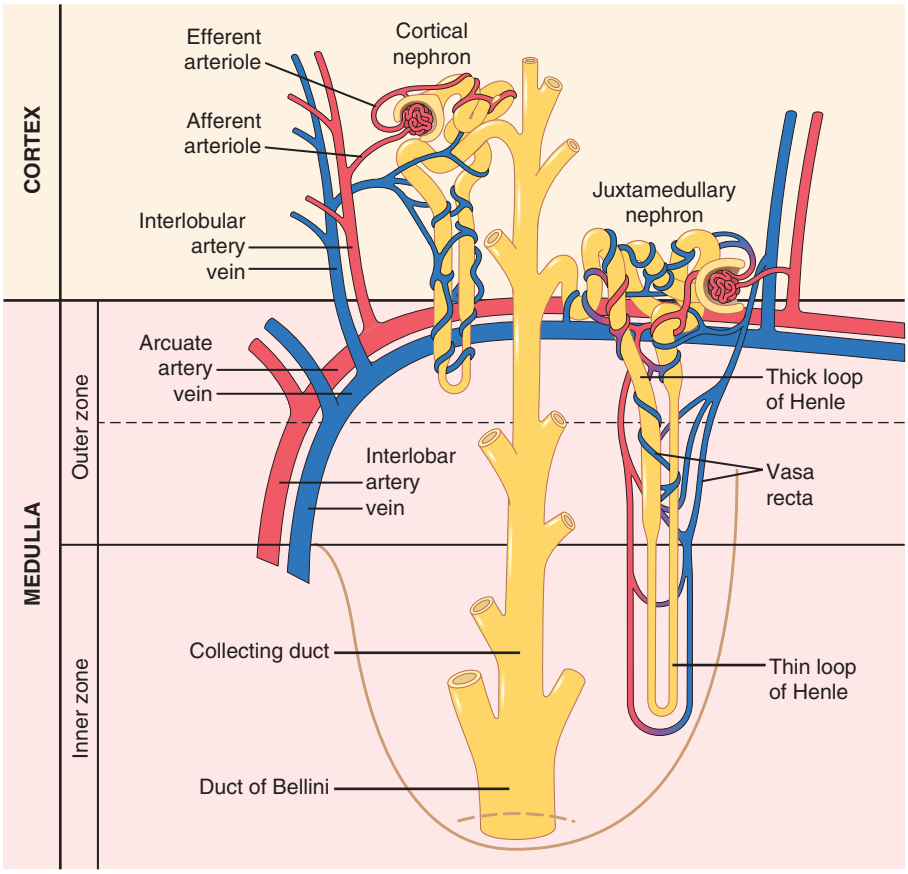


Figure 26.5 Schematic of relationships between blood vessels and tubular structures and differences between cortical and juxtamedullary nephrons.

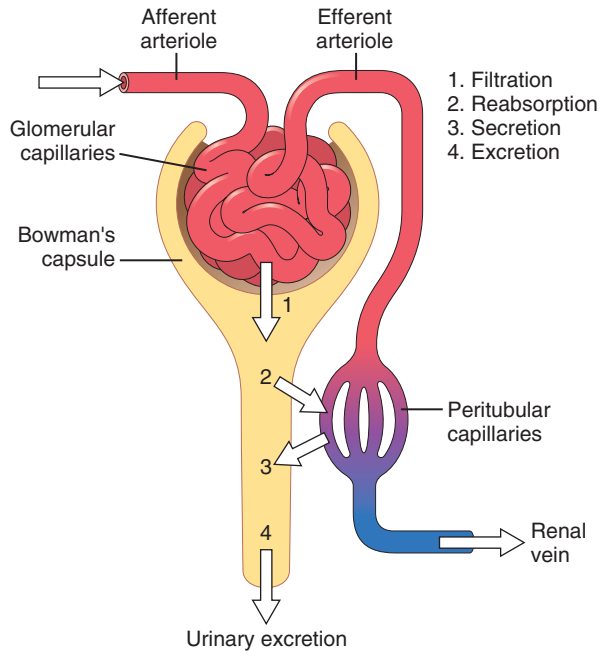


Figure 26.6 Basic kidney processes that determine the composition of the urine. The urinary excretion rate of a substance is equal to the rate at which the substance is filtered minus its reabsorption rate plus the rate at which it is secreted from the peritubular capillary blood into the tubules.

tubules into the blood; and (3) secretion of substances from the blood into the renal tubules as follows:

Urinary excretion rate
= Filtration rate − Reabsorption rate
+ Secretion rate

Urine formation begins when a large amount of fluid that is virtually free of protein is filtered from the glomerular capillaries into Bowman's capsule. Most substances in the plasma, except for proteins, are freely filtered, so their concentration in the glomerular filtrate in Bowman's capsule is almost the same as in the plasma. As filtered fluid leaves Bowman's capsule and passes through the tubules, it is modified by reabsorption of water and specific solutes back into the blood or by secretion of other substances from the peritubular capillaries into the tubules.

Fig. 26.7 shows the renal handling of four hypothetical substances. The substance shown in panel A is freely filtered by the glomerular capillaries but is neither reabsorbed nor secreted. Therefore, its excretion rate is equal to the rate at which it was filtered. Certain waste products in the body, such as creatinine, are handled by the kidneys in this manner, allowing excretion of essentially all that is filtered.

In panel B, the substance is freely filtered but is also partly reabsorbed from the tubules back into the blood.

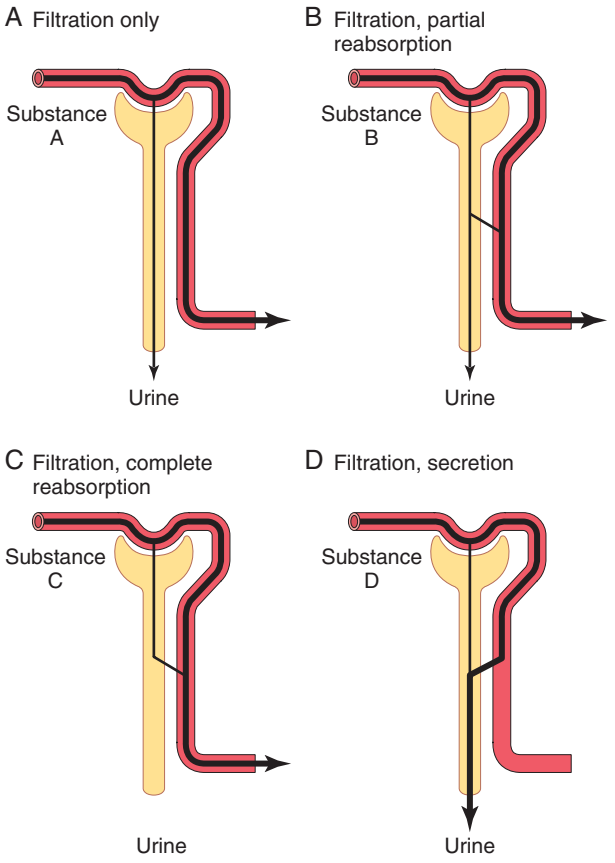


Figure 26.7 Renal handling of four hypothetical substances. (A) The substance is freely filtered but not reabsorbed. (B) The substance is freely filtered, but part of the filtered load is reabsorbed back in the blood. (C) The substance is freely filtered but is not excreted in the urine because all the filtered substance is reabsorbed from the tubules into the blood. (D) The substance is freely filtered and is not reabsorbed but is secreted from the peritubular capillary blood into the renal tubules.

Therefore, the rate of urinary excretion is less than the rate of filtration at the glomerular capillaries. In this case, the excretion rate is calculated as the filtration rate minus the reabsorption rate. This pattern is typical for many of the electrolytes of the body, such as sodium and chloride ions.

In panel C, the substance is freely filtered at the glomerular capillaries but is not excreted into the urine because all the filtered substance is reabsorbed from the tubules back into the blood. This pattern normally occurs for some of the nutritional substances in the blood, such as amino acids and glucose, allowing them to be conserved in the body fluids.

The substance in panel D is freely filtered at the glomerular capillaries and is not reabsorbed, but additional quantities of this substance are secreted from the peritubular capillary blood into the renal tubules. This pattern often occurs for organic acids and bases, permitting them to be cleared from the blood rapidly and excreted in large amounts in the urine. The excretion rate in this case is calculated as filtration rate plus tubular secretion rate.

For each substance in the plasma, a particular combination of filtration, reabsorption, and secretion occurs. The rate at which each substance is excreted in the urine depends on the relative rates of these three basic renal processes.

FILTRATION, REABSORPTION, AND SECRETION OF DIFFERENT SUBSTANCES

In general, tubular reabsorption is quantitatively more important than tubular secretion in the formation of urine, but secretion plays an important role in determining the amounts of potassium and hydrogen ions and a few other substances that are excreted in the urine. Most substances that must be cleared from the blood, especially the end products of metabolism, such as urea, creatinine, uric acid, and urates, are poorly reabsorbed and are therefore excreted in large amounts in the urine. Certain foreign substances and drugs are also poorly reabsorbed but, in addition, are secreted from the blood into the tubules, so their excretion rates are high. Conversely, electrolytes, such as sodium ions, chloride ions, and bicarbonate ions, are highly reabsorbed, so only small amounts appear in the urine. Certain nutritional substances, such as amino acids and glucose, are completely reabsorbed from the tubules and do not appear in the urine, even though large amounts are filtered by the glomerular capillaries.

Each of the processes—glomerular filtration, tubular reabsorption, and tubular secretion—is regulated according to the needs of the body. For example, when there is excess sodium in the body, the rate at which sodium is filtered usually increases slightly, and a smaller fraction of the filtered sodium is reabsorbed, causing increased urinary excretion of sodium. When there is excess potassium, increased urinary excretion of potassium is achieved mainly by increasing tubular secretion of potassium. As we discuss in the remaining chapters of this section, various combinations of changes in the rates of glomerular filtration, tubular reabsorption, and tubular secretion are used to regulate excretion of different substances.

For most substances, the rates of filtration and reabsorption are extremely large relative to the rates of excretion. Therefore, even slight changes of filtration or reabsorption can lead to relatively large changes in renal excretion. For example, an increase in the glomerular filtration rate (GFR) of only 10% (from 180 to 198 L/day) would raise the urine volume by 13-fold (from 1.5 to 19.5 L/day) if tubular reabsorption remained constant. In reality, changes in glomerular filtration, tubular reabsorption, and tubular secretion usually act in a coordinated manner to produce the necessary changes in renal excretion.

Why Are Large Amounts of Solutes Filtered and Then Reabsorbed By the Kidneys? One might question the wisdom of filtering such large amounts of water and solutes and then reabsorbing most of these substances. The high rate of tubular reabsorption required to recover most of the electrolytes, glucose, amino acids, and many other solutes filtered is metabolically expensive; approximately